EKURHULENI WEST TVET COLLEGE

Facilitator	**RAPHATELO MD**
Programme/Course	**ICT**
Module	**COMPUTER PROGRAMMING L2**
Module Code	**CPROGL2**
Assessment Type	**PRACTICAL FORMATIVE ASSESSMENT Memorandum**
Topic	**Customer Shopping Bill – Python**
Instructions	**1. Answer ALL questions.** **2. Use Python 3 in Thonny.** **3. Submit your IPO chart, Python file and test record.**
Duration	**60 minutes**

Question 1

1.1

INPUT	PROCESS	OUTPUT
Customer name Quantity Unit price	Subtotal = quantity × price If subtotal ≥ R50.00: discount = subtotal × 0.10 Otherwise: discount = 0 Final = subtotal − discount	Customer name Subtotal (R) Discount (R) Final amount (R)

Question	Model answer / observable evidence	Marks
1.2	customer_name = input("Enter customer name: ") quantity = int(input("Enter number of notebooks: ")) price = float(input("Enter price per notebook (R): "))	(5)

Question	Model answer / observable evidence	Marks

1.3	subtotal = quantity * price
	if subtotal >= 50:
	  discount = subtotal * 0.10
	else:
	  discount = 0.0
	final_amount = subtotal - discount
		(4)

Question	Model answer / observable evidence	Marks
1.4	print("\n--- CUSTOMER BILL ---") print("Customer:", customer_name) print(f"Subtotal: R{subtotal:.2f}") print(f"Discount: R{discount:.2f}") print(f"Final amount: R{final_amount:.2f}")	(4)

Question	Model answer / observable evidence	Marks
1.5	# Stationery Shop Billing Program # Calculates the bill and applies a 10% discount for subtotals of R50 or more	(1)

1.6 (1)

Test	Customer	Quantity	Price (R)	Subtotal (R)	Discount (R)	Final (R)
1	Lebo	5	20.00	100.00	10.00	90.00
2	Thabo	2	15.50	31.00	0.00	31.00
3	Naledi	2	25.00	50.00	5.00	45.00

Question 2

Question	Model answer / observable evidence	Marks
2.1	input() returns text (strings), so multiplying quantity by price raises a TypeError. Correct: quantity = int(input("Enter quantity: ")) price = float(input("Enter price (R): "))	(2)

Question	Model answer / observable evidence	Marks

2.2	The result is 20 because multiplication (5 × 2 = 10) happens before addition (10 + 10 = 20).	(2)

Question	Model answer / observable evidence	Marks
2.3	result = (10 + 5) * 2	(1)

MARK SUMMARY

Question	Assessment criterion	Marks
1.1	IPO chart	4
1.2	Input and conversion	5
1.3	Calculations and condition	4
1.4	Output	4
1.5	Documentation	1
1.6	Testing	2
2.1–2.3	Reasoning	5
	TOTAL	25